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Amygdala Cell Types in Humans and Rodents Across Classical and Molecular Dimensions

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Introduction and amygdala subdivision background

The amygdala is not one uniform structure but a complex made of many nuclei, often grouped into basolateral/corticobasal, central, medial, superficial/cortical-like, and intercalated components. Across rodents, primates, and humans, this broad plan is largely conserved, and it sets up the main distinction between cortical-like glutamatergic regions and striatum-like, mostly GABAergic regions. (10 sources)

The amygdala is best understood as an amygdaloid complex rather than a single nucleus. Depending on the naming system, it includes roughly a dozen or more subregions that are commonly grouped into major nuclear divisions on the basis of cytoarchitecture, chemistry, and connectivity. A widely used scheme separates basolateral or deep nuclei, superficial or cortical-like nuclei, and centromedial nuclei, with intercalated cell masses and nearby accessory areas often treated as additional components. In rodents, the deep/basolateral group includes the lateral, basal, and accessory basal nuclei; the superficial group includes the cortical nuclei and nucleus of the lateral olfactory tract; and the centromedial group includes the medial and central nuclei. Other descriptions similarly divide the amygdala into basolateral, central, medial, and intercalated compartments, with more than 13 subregions recognized in some accounts. (Veinante et al., 2013) (Pineda et al., 2021) (Aerts et al., 2021) (Hu et al., 2022)

A notable organizing idea is that these subdivisions fall into two broad macrostructural types. The corticobasolateral complex, which includes the basolateral nuclei and often the cortical-like superficial nuclei, has cortex-like features and contains cell types that resemble cortical pyramidal and nonpyramidal neurons. By contrast, the centromedial complex shows striatopallidal or striatum-like organization. This distinction is useful because it predicts the main cell classes later discussed: basolateral regions are dominated by glutamatergic principal neurons with local GABAergic interneurons, whereas the central and much of the medial amygdala are largely composed of inhibitory GABAergic neurons. (Bzdok et al., 2012) (Wilson et al., 2015) (Veinante et al., 2013) (Hu et al., 2022)

Within this broad map, the basolateral nuclear group is usually defined as the lateral, basolateral/basal, and basomedial/accessory basal nuclei, although exact names differ by atlas and species. Human and non-human primate studies often further subdivide the basal nucleus into magnocellular, intermediate, and parvicellular parts, while rodent studies more often split it into anterior and posterior divisions. This means that some cell-type names are conserved across species at the major nucleus level, but finer subnuclear labels can differ. (Aerts et al., 2021) (Zhu et al., 2025)

Across mammals, this overall nuclear layout appears reasonably conserved. Comparative work has reported homology of major amygdala nuclei across rats, cats, monkeys, and humans, and newer summaries likewise describe the amygdala as a conserved constellation of nuclei and cell types in rodents, primates, and humans. Quantitative scaling differs by nucleus, but the shared basic plan supports cross-species comparison when discussing classical and molecularly defined amygdala cell types. (Bzdok et al., 2012) (Porcaro et al., 2023) (Chareyron et al., 2011)

One final terminology point is important for the rest of the summary: some authors place the superficial nuclei together with the basolateral nuclei under a broader corticobasal or cortical-like amygdala, while others emphasize a simpler split between excitatory-output nuclei and inhibitory-output nuclei. In that framing, the basolateral complex and cortical nuclei are mainly excitatory-output regions, whereas central and much of medial amygdala circuitry is inhibitory-output and more striatal in character. Intercalated cell masses stand apart as small clusters of densely packed GABAergic neurons positioned between major nuclei. (Veinante et al., 2013) (Carney et al., 2010) (Wilson et al., 2015)

Evidence

(Veinante et al., 2013)

The amygdala between sensation and affect: a role in pain

> *"The amygdala is an heterogeneous structure, grouping around a dozen of nuclei depending on the considered nomenclature, these nuclei being classically clustered in 4 groups: superficial, basolateral, central and medial (Figure 1) [7,8]. The first two groups constitute the corticobasal amygdala that shows cortical-like characteristics [8]. Its main nuclei, the lateral and the basolateral nuclei are known as the basolateral amygdala (BLA) and mostly contain projection neurons of pyramidal type, synthesizing glutamate"*

> *".These last two groups display an organization that is similar to the one of the basal ganglia, with striatopallidal-like neuronal morphology (Alheid, 1995)(Cassell et al., 1999)"*

> *".among them the intercalated cell masses which are small clusters of densely packed GABAergic neurons (Palomares-Castillo et al., 2012)."*

(Pineda et al., 2021)

Extrahypothalamic Control of Energy Balance and Its Connection with Reproduction: Roles of the Amygdala

> *"In rodents, these different nuclei are divided into three main groups: (i) the deep or basolateral group, which includes the lateral, basal, and accessory basal nuclei; (ii) the superficial or cortical group, which includes the cortical nuclei and nucleus of the lateral olfactory tract; and (iii) the centromedial group, which includes the medial and central nuclei. In addition, other accessory nuclei, including the intercalated cell masses and the amygdalo-hippocampal area, have been also described (Pabba, 2013)(Davis et al., 2000)(Sah et al., 2003)."*

(Aerts et al., 2021)

Novel Perspectives on the Development of the Amygdala in Rodents

> *"On the basis of its cytoarchitecture, neurochemistry and connectivity, the amygdaloid complex was originally subdivided into a basolateral complex (BLC), a centromedial complex and a superficial cortical-like complex (Johnston, 1923)(Alheid, 1995)(McDonald, 1998)(Sah et al., 2003). Here, the BLC comprises the lateral amygdala (LA), the basolateral amygdala (BLA), and the basomedial amygdala (BM). The latter two nuclei are sometimes also referred to as the basal amygdala (BA) and the accessory basal amygdala (AB), respectively. The cortical-like nuclei or superficial nuclei include the nucleus of the lateral olfactory tract (nLOT), bed nucleus of the accessory olfactory tract (BAOT) and the cortical nuclei (CoA, anterior and posterior, ACo and PCo, resp). The centromedial nuclei consist of the central amygdala (CA), medial amygdala (MA), and the intra-amygdaloid part of the bed nucleus of the stria terminalis (BSTia)."*

(Hu et al., 2022)

New Insights into the Pivotal Role of the Amygdala in Inflammation-Related Depression and Anxiety Disorder

> *"The amygdala complex contains more than 13 subregions, which can be distinguished according to their neuronal morphology, unique connectivity, physiology, and expression of specific molecular markers. It is usually divided into the basolateral amygdala (BLA), central amygdala (CeA), medial amygdala (MeA), and intercalated cell clusters [78]"*

> *".Developmentally, the BLA is a cortical-like structure consisting of 80% glutamatergic principal neurons and approximately 20%*

GABAergic inhibitory neurons (Carlsen et al., 1988)(Keifer et al., 2015). In contrast, CeA is a striatum-like structure that primarily comprises GABAergic inhibitory neurons (McDonald, 1982)."

(Bzdok et al., 2012)

[An Investigation of the Structural, Connectional, and Functional Subspecialization in the Human Amygdala]()

> *"Comprehensive studies in rats, cats, and monkeys revealed reasonable homology between the amygdala nuclei regarding microanatomy and connectivity [Price et al., 1987]. Laterobasal, centromedial, and superficial nuclei groups were isolated and thoroughly studied in these three species (McDonald, 1998). Cytoarchitectonically, the laterobasal nuclei group resembles pyramidal and nonpyramidal neurons of the cerebral cortex (Hall, 1972), while the architecture of the superficial nuclei especially resembles that of the olfactory cortex [McDonald, 1992]. Conversely, the centromedial nuclei group does not resemble the cerebral cortex but shows similarities with cell types in the striatum [Heimer and Van Hoesen, 2006;McDonald,"*

(Wilson et al., 2015)

[Stress as a one-armed bandit: Differential effects of stress paradigms on the morphology, neurochemistry and behavior in the rodent amygdala]()

> *"the amygdalar nuclei can be grouped into two major macrostructures. First is the corticobasolateral nuclei, whose cell types resemble those of the cerebral cortex, with glutamatergic, calcium-calmodulin dependent (CaM) kinase-positive principal cells that bear morphological resemblance to cortical pyramidal neurons (McDonald, 1996). The activity of these neurons is coordinated, synchronized, and otherwise modulated by a variety of different inhibitory, GABAergic-interneurons, which can be classified according to their expression of various calcium-binding proteins as well as by morphological and connectional features. The second major amygdalar macrostructure is the centromedial complex, whose cell types resemble those of the striatopallidal region."*

(Zhu et al., 2025)

[Brain-wide connections of the parvicellular subdivision of the basolateral and basomedial amygdaloid nuclei in the rats]()

> *"The basolateral nuclear group contains three major nuclei, and these include lateral (La), basolateral (BL) (or basal [B]) and basomedial (BM) (or accessory basal [AB]) nuclei depending on different nomenclature (e.g., De Olmos et al., 1985;(Aggleton, 1992)(Pitkänen et al., 1998)(Fudge et al., 2002)Paxinos and Watson, 2007;(Aggleton, 1992)"*

> *".In human and non-human primate, the BL (or B) is further parcellated into magnocellular (Bmc), intermediate (Bi) and parvicellular (Bpc) subdivisions (Aggleton, 1992)(Fudge et al., 2002)(Aggleton, 1992)(McHale et al., 2021) or into dorsal (BLD), intermediate (BLi) and ventral (BLV) subdivisions, respectively (e.g., Paxinos et al., 2009;Ding et al., 2016;Ding et al., 2022). For consistent use of these terminology, we use BLmc, BLi and BLpc in this study. In rodents, however, parcellation of the BL is inconsistent among different research groups. For example, the BL in rats and mice is often subdivided into the anterior and posterior parts (BLA and BLP, respectively) by many authors (De Olmos et al., 1985;Swanson, 1992;Paxinos and Watson, 2007;Wang et al., 2020)."*

(Porcaro et al., 2023)

[Seeking the Amygdala: Novel Use of Diffusion Tensor Imaging to Delineate the Basolateral Amygdala]()

> *"The amygdala itself is a constellation of nuclei and cell types that are remarkably conserved among mammalian species, including rat, monkey and human (Chareyron et al., 2011). The amygdala is primarily composed of the basolateral (BLA) amygdala (lateral, basal, and accessory basal nuclei), and the central, medial and cortical nuclei, which have been termed the amygdaloid complex (Davis et al., 2000)(Pitkänen et al., 1997)."*

(Chareyron et al., 2011)

[Stereological Analysis of the Rat and Monkey Amygdala]()

> *"The amygdala is part of a neural network that contributes to the regulation of emotional behaviors. Rodents, especially rats, are used extensively as model organisms to decipher the functions of specific amygdala nuclei, in particular in relation to fear and emotional learning. Analysis of the role of the nonhuman primate amygdala in these functions has lagged work in the rodent but provides evidence for conservation of basic functions across species. Here we provide quantitative information regarding the morphological characteristics of the main amygdala nuclei in rats and monkeys, including neuron and glial cell numbers, neuronal soma size, and individual nuclei volumes. The volumes of the lateral, basal, and accessory basal nuclei were, respectively, 32, 39, and 39 times larger in monkeys than in rats. In contrast, the central and medial nuclei were only 8 and 4 times larger in monkeys than in rats. The numbers of neurons in the lateral, basal, and accessory basal nuclei were 14, 11, and 16 times greater in monkeys than in rats, whereas the numbers of neurons in the central and medial nuclei were only 2.3 and 1.5 times greater in monkeys than in rats. Neuron density was between 2.4 and 3.7 times lower in monkeys than in rats, whereas glial density was only between 1.1 and 1.7 times lower in monkeys than in rats. We compare our data in rats and monkeys with those previously published in humans and discuss the theoretical and functional implications that derive from our quantitative structural findings. J. Comp. Neurol. 519:3218–3239, 2011. © 2011 Wiley-Liss, Inc."*

(Carney et al., 2010)

[Sonic hedgehog expressing and responding cells generate neuronal diversity in the medial amygdala]

> *"The amygdala is considered in terms of two functional subdivisions, classified according to whether the major output projection of each nucleus is excitatory or inhibitory. Based on this classification, the nuclei of the basolateral complex and cortical amygdalar nuclei, which have an excitatory output"*
>
> *"the cortical and medial nuclei are broadly considered similar to the striatum as their output projections are primarily inhibitory."*

Classical neuron classes across amygdala subdivisions

Across amygdala nuclei, the main classical split is between excitatory principal/projection neurons and inhibitory local neurons. Basolateral and cortical-like regions are dominated by glutamatergic, often pyramidal-like spiny cells, while central and much of medial amygdala are dominated by GABAergic neurons, with intercalated cell masses forming small inhibitory clusters. (11 sources)

A useful classical framework is that most amygdala subdivisions can be described with the same broad cell-class logic, even though the exact mix differs by nucleus. In the corticobasolateral amygdala, especially the basolateral complex, the dominant cells are principal or projection neurons that are glutamatergic, spiny, and often pyramidal-like, while the main alternative class is nonpyramidal, spine-poor GABAergic interneurons. This two-class scheme has been a notable organizing principle in the field and is described in both older anatomical work and recent reviews. (McDonald et al., 2012) (Pare et al., 1996) (McDonald, 2024)

In the basolateral amygdala, these glutamatergic principal neurons make up about 70-90% of neurons depending on the nucleus and method, with GABAergic interneurons forming the minority. They are usually described as spiny multipolar or pyramidal-like cells with long axons and local collaterals, much like cortical projection neurons. By contrast, the inhibitory cells are morphologically diverse, often aspiny or sparsely spiny, and include stellate, bipolar, bitufted, tufted, triangular, oval, round, and multipolar forms. (Ignacio et al., 2014) (Pare et al., 1996) (Perumal et al., 2021) (Hajos, 2021) (Vereczki et al., 2021) (McDonald, 2024)

Classical interneuron categories in the corticobasolateral amygdala were first defined by marker content and morphology. A commonly cited scheme identifies at least four GABAergic nonpyramidal subgroups: parvalbumin/calbindin-positive cells, somatostatin/calbindin-positive cells, large multipolar cholecystokinin-positive cells, and small bipolar or bitufted cells that often coexpress calretinin, cholecystokinin, and VIP. More recent work shows that the lateral and basal nuclei also contain familiar cortical inhibitory classes such as axo-axonic cells, PV basket cells, CCK basket cells, somatostatin dendrite-targeting cells, NPY neurogliaform cells, VIP/calretinin interneuron-selective interneurons, and some GABAergic projection neurons. (McDonald et al., 2012) (Hajos, 2021) (Vereczki et al., 2021) (Andrasi et al., 2017)

Outside the basolateral group, the balance shifts. The central amygdala is classically described as being composed mainly of GABAergic neurons rather than glutamatergic principal cells, and the medial amygdala is also largely inhibitory, though some rodent glutamatergic projection neurons are present there. In broad terms, principal neurons are exclusively glutamatergic in BLA, cortical amygdala, and basomedial amygdala, exclusively GABAergic in central amygdala, and predominantly GABAergic in medial amygdala. (Ignacio et al., 2014) (Sarowar et al., 2020) (Zhang et al., 2021) (Raudales et al., 2024)

Intercalated cell masses form another classical category. Rather than a large principal-cell field, they are small clusters of densely packed inhibitory neurons located between the basolateral and central nuclei, and are often noted for dopamine D1 receptor and μ-opioid receptor expression. (Sarowar et al., 2020) (Zhang et al., 2021)

So, if one asks for the classical cell types across the amygdala as a whole, the shortest accurate answer is: glutamatergic spiny principal/projection neurons in the cortical-like nuclei; diverse GABAergic nonpyramidal interneurons in those same nuclei; mostly GABAergic projection-type neurons in the central nucleus and much of the medial nucleus; and small inhibitory intercalated cell clusters at the borders between major nuclei. (McDonald et al., 2012) (Pare et al., 1996) (Zhang et al., 2021) (Raudales et al., 2024)

Evidence

(McDonald et al., 2012)

[Subpopulations of somatostatin-immunoreactive non-pyramidal neurons in the amygdala and adjacent external capsule project to the basal forebrain: evidence for the existence of GABAergic projection neurons in the cortical nuclei and basolateral nuclear complex]

"The cell types in all of these amygdalar nuclei are similar, but they have been studied primarily in the basolateral amygdala. The principal neurons in the CBL are pyramidal-like projection neurons with spiny dendrites that utilize glutamate as an excitatory neurotransmitter, whereas most non-pyramidal neurons in the CBL are spine-sparse interneurons that utilize GABA as an inhibitory neurotransmitter (McDonald, 1982(McDonald, , 1985(McDonald, , 1992a(McDonald, ,b, 1996(McDonald, , 2003Millhouse and DeOlmos, 1983;Fuller et al., 1987;Carlsen and Heimer, 1988;McDonald and Augustine, 1993). Dual-labeling immunohistochemical studies in the basolateral amygdala suggest that the CBL contains at least four distinct subpopulations of GABAergic non-pyramidal neurons that can be distinguished on the basis of their content of calcium-binding proteins and peptides. These subpopulations are: (1) PV+/CB+ neurons, (2) SOM+/CB+ neurons, (3) large multipolar cholecystokinin+ neurons that are often CB+, and (4) small bipolar and bitufted interneurons that exhibit extensive colocalization of calretinin, cholecystokinin, and vasoactive intestinal peptide (Kemppainen and Pitkänen, 2000;McDonald and Betette, 2001;Mascagni, 2001, 2002;"

(Pare et al., 1996)

[Projection Cells and Interneurons of the Lateral and Basolateral Amygdala: Distinct Firing Patterns and Differential Relation to Theta and Delta Rhythms in Conscious Cats]

"Golgi observations indicate that the BL complex contains two main cell types (Hall, 1972;Tomb61 and Szafranska-Kosmal, 1972;Kamal and Tombol, 1975) (for review, see McDonald, 1992). The most common type is a spiny multipolar neuron that often has a dominant dendrite giving it a pyramidal appearance. These cells are excitatory projection neurons because they have long axons that emit numerous collaterals and use an excitatory amino acid as transmitter (Christie et al., 1987;Fuller et al., 1987;(Smith et al., 1994)(Smith et al., 1994). The second cell type consists of a morphologically heterogeneous group of sparsely spiny neurons that have a locally ramifying axon, are immunopositive for GABA (McDonald, 1985;(McDonald et al., 1993)Pare and Smith, 1993) and thus constitute a class of inhibitory interneurons."

(McDonald, 2024)

[Functional neuroanatomy of basal forebrain projections to the basolateral amygdala: transmitters, receptors, and neuronal subpopulations]

"There are two major neuronal classes in the BNC: pyramidal neurons (PNs) and nonpyramidal neurons (NPNs)"

". BNC PNs constitute about 80-90% of BNC neurons and are the main projection neurons of the BNC (McDonald, 1992b).NPNs in the BNC have aspiny or spine-sparse dendrites and closely resemble NPNs of the cortex (Hall, 1972;McDonald, 1982;Millhouse & DeOlmos, 1983).Since the great majority of NPNs are interneurons (INs) (McDonald, 2020), the latter term will be used to denote these neurons in the remainder of this review."

(Ignacio et al., 2014)

[Effects of Acute Prenatal Exposure to Ethanol on microRNA Expression are Ameliorated by Social Enrichment]

"Neuronal types differ considerably among the subdivisions of the amygdala (Sah et al., 2003). In the basolateral group, approximately 70% of neurons are thought to be glutamatergic (pyramidal, spiny, or class I neurons). This division also contains interneurons such as GABAergic nonspiny stellate cells of the cortex (called S cells, stellate, or class II neurons). In contrast, within the central nucleus, the majority of cells are thought to be GABAergic."

(Perumal et al., 2021)

[Inhibitory Circuits in the Basolateral Amygdala in Aversive Learning and Memory]

"The BLA is a cortical like structure, and similar to other cortical regions, local microcircuits are formed by excitatory glutamatergic neurons that make [illegible] 80%of total neuronal population and GABAergic interneurons (Sah et al., 2003)(Tovote et al., 2015)"

".In the BLA, many glutamatergic neurons resemble their cortical counterparts with a classical pyramidal soma, thicker apical dendrite and a spiny dendritic tree (McDonald, 1982). However, glutamatergic neurons in the lateral amygdala often lack pyramidal shaped soma or clear basal and apical dendrites as in the cortex. As glutamatergic neurons constitute the main neuronal subtype in the amygdala, they are referred as principal neurons (PNs) (Faber et al., 2001)(Sah et al., 2003). In contrast, GABAergic neurons comprise of a diverse population of cells with different somatic morphologies such as triangular, oval, round or square shaped somata with bipolar or tuft dendrites that are either aspiny or sparsely spiny (McDonald, 1982)."

(Hajos, 2021)

[Interneuron Types and Their Circuits in the Basolateral Amygdala]

"Amygdala is the brain region where at least 13 different nuclei are defined with typical neuron types, developmental origin, and connectivity patterns (Pitkänen et al., 1997)(Swanson, 2003)"

"Based on the cell types, their connectivity features, and developmental characteristics, the BLA is a cortical structure. Accordingly, glutamatergic excitatory projection cells expressing vesicular glutamate transporter type 1 (VGluT1; (Andrási et al., 2017) are the most numerous neurons in this amygdala region (80-85%; (Vereczki et al., 2021). The dendrites of the principal cells (PC) are densely decorated with spines and their axon arborizes within the nucleus, giving rise to local collaterals, but they also project to other amygdala regions and remote cortical and/or subcortical areas (Sah et al., 2003)."

(Vereczki et al., 2021)

Total Number and Ratio of GABAergic Neuron Types in the Mouse Lateral and Basal Amygdala

"we estimated that the following cell types together compose the vast majority of GABAergic cells in the LA and BA: axo-axonic cells (5.5%-6%), basket cells expressing parvalbumin (17%-20%) or cholecystokinin (7%-9%), dendrite-targeting inhibitory cells expressing somatostatin (10%-16%), NPY-containing neurogliaform cells (14%-15%), VIP and/or calretinin-expressing interneuron-selective interneurons (29%-38%), and GABAergic projection neurons expressing somatostatin and neuronal nitric oxide synthase (5.5%-8%). Our results show that these amygdalar nuclei contain all major GABAergic neuron types as found in other cortical regions."

(Andrasi et al., 2017)

Differential excitatory control of 2 parallel basket cell networks in amygdala microcircuits

"Information processing in neural networks depends on the connectivity among excitatory and inhibitory neurons. The presence of parallel, distinctly controlled local circuits within a cortical network may ensure an effective and dynamic regulation of microcircuit function. By applying a combination of optogenetics, electrophysiological recordings, and high resolution microscopic techniques, we uncovered the organizing principles of synaptic communication between principal neurons and basket cells in the basal nucleus of the amygdala. In this cortical structure, known to be critical for emotional memory formation, we revealed the presence of 2 parallel basket cell networks expressing either parvalbumin or cholecystokinin. While the 2 basket cell types are mutually interconnected within their own category via synapses and gap junctions, they avoid innervating each other, but form synaptic contacts with axo-axonic cells. Importantly, both basket cell types have the similar potency to control principal neuron spiking, but they receive excitatory input from principal neurons with entirely diverse features. This distinct feedback synaptic excitation enables a markedly different recruitment of the 2 basket cell types upon the activation of local principal neurons. Our data suggest fundamentally different functions for the 2 parallel basket cell networks in circuit operations in the amygdala."

(Sarowar et al., 2020)

Rho GTPases in the Amygdala—A Switch for Fears?

"The majority of BLA neurons are spiny glutamatergic neurons (with a minority of GABAergic interneurons) (Spampanato et al., 2011). CEl and CEm mainly contain GABAergic neurons. The ICMs are small cell clusters and consist of more dopamine type-1 and μ-opioid-receptor expressing cells (Poulin et al., 2008)."

(Zhang et al., 2021)

Amygdala Circuit Substrates for Stress Adaptation and Adversity.

"Because of differing developmental origins, certain amygdala nuclei exhibit a distinct neuronal composition (Sah et al., 2003)(Davis et al., 2000). The BLA has a more cortical-like profile, primarily containing excitatory neurons, whereas the CeA and MeA have a striatal-like composition of largely inhibitory neurons (Sah et al., 2003)(McDonald, 2003)(McDonald, 1982). Another discrete set of inhibitory nuclei, known as intercalated cell clusters, is located at the intersection of the BLA and CeA (Marowsky et al., 2005)."

(Raudales et al., 2024)

Specific and comprehensive genetic targeting reveals brain-wide distribution and synaptic input patterns of GABAergic axo-axonic interneurons

"The amygdaloid complex includes over a dozen nuclei and can be segregated into five groups (Beyeler and Dabrowska, 2020): (1) the BLA divided into a dorsal section (lateral amygdala, LA) and basal section (basal amygdala, BA), (2) the basomedial amygdala (BMA), (3) the central amygdala (CeA) further splits into medial, lateral, and central sections (CeM, CeL, and CeC), (4) the medial amygdala (MeA), and (5) the cortical amygdala (CoA)"

".Within the amygdala nuclei, PNs are exclusively glutamatergic in BLA, CoA, BMA, exclusively GABAergic in CeA, and predominantly GABAergic in MeA and BST.In rodents, there is also a population of glutamatergic pyramidal neurons (GLU PNs, derived from third ventricle neuroepithelium) that populates the BST, MeA, and hypothalamus (García-Moreno et al., 2010)(Huilgol et al., 2016)."

Basolateral amygdala and corticobasal cell types

In the basolateral and related corticobasal amygdala, the main cells are glutamatergic pyramidal-like principal neurons plus *many named GABAergic interneuron types. Rodent work shows that these nuclei contain most of the same major inhibitory classes seen in cortex, alongside several physiological firing classes. (7 sources)*

- **Principal or projection neurons**
- The dominant cell type in the basolateral amygdala (BLA; including lateral and basal nuclei) is the **glutamatergic principal neuron**, usually described as **pyramidal-like** and making up roughly **85-90%** of neurons. These cells resemble cortical and hippocampal pyramidal neurons and are the main long-range projection cells. (Woodruff et al., 2007) (Washburn et al., 1992)
- In physiological studies, most recorded BLA neurons were identified as **pyramidal cells**, and biocytin fills showed morphologies ranging from **pyramidal to stellate cell bodies**. (Niimi et al., 2012) (Washburn et al., 1992)
- **Broad physiological classes in BLA recordings**
- Classical rodent slice studies separated BLA neurons into **pyramidal cells**, **late-firing neurons**, and **fast-firing or first-firing neurons**. The latter two classes are commonly treated as inhibitory interneuron-like groups, while pyramidal cells are the main projection class. (Washburn et al., 1992) (Niimi et al., 2012)
- **Parvalbumin-positive interneurons**
- A notable inhibitory class is the **parvalbumin-positive (PV+) interneuron** group. This includes **PV basket cells**, which target principal-cell somata and proximal dendrites, and **PV axo-axonic cells**, which target the axon initial segment. (Woodruff et al., 2007) (Vereczki et al., 2016) (Vereczki et al., 2021)
- More specifically, estimates for lateral and basal nuclei suggest **PV basket cells** make up about **17-20%** of GABAergic cells, while **axo-axonic cells** make up about **5.5-6%**. (Vereczki et al., 2021)
- **Cholecystokinin-positive basket cells**
- Another major perisomatic inhibitory class is the **CCK basket cell**, often marked by **cholecystokinin** and **CB1 cannabinoid receptor** expression. These cells innervate principal-cell somata and proximal dendrites alongside PV basket cells. (Woodruff et al., 2007) (Vereczki et al., 2016) (Rovira-Esteban et al., 2019)
- In recent estimates, **CCK basket cells** account for about **7-9%** of GABAergic cells in the lateral and basal nuclei. (Vereczki et al., 2021)
- **Somatostatin dendrite-targeting interneurons**
- The BLA also contains **somatostatin-positive (SOM+) interneurons**, often described as **dendrite-targeting inhibitory cells**. Earlier summaries also grouped many of these with **somatostatin/calbindin-positive** interneurons. (Woodruff et al., 2007) (Bienvenu et al., 2012) (Vereczki et al., 2021)
- These **SOM dendrite-targeting cells** were estimated at about **10-16%** of GABAergic neurons in LA/BA. (Vereczki et al., 2021)
- **Calbindin-positive dendrite-targeting interneurons**
- A related class described in the BLA is the **calbindin-positive interneuron targeting dendrites**. This emphasizes that dendrite-targeting inhibition in BLA can be split by marker content as well as by target domain. (Bienvenu et al., 2012)
- **Neurogliaform cells**
- The BLA contains **NPY-containing neurogliaform cells**, a notable cortical-like inhibitory type also found in amygdala. (Vereczki et al., 2021)
- Experimental classifications have also directly identified **neurogliaform cells** among BLA interneurons.

(Rovira-Esteban et al., 2019)

- **VIP/calretinin interneuron-selective interneurons**
- Another named class is **VIP- and/or calretinin-expressing interneuron-selective interneurons**, which mainly target other interneurons rather than principal cells. (Vereczki et al., 2021)
- Older marker-based schemes also noted **small bipolar or bitufted cells** that often coexpress **calretinin, CCK, and VIP**, which likely overlaps with this interneuron-selective group. (Bienvenu et al., 2012)
- **GABAergic projection neurons**
- Although most inhibitory cells in BLA are local interneurons, there is also a distinct class of **GABAergic projection neurons**. These have been described as a separate projecting interneuron-like type and, in more recent taxonomy, as **somatostatin- and neuronal nitric oxide synthase-expressing GABAergic projection neurons**. (Bienvenu et al., 2012) (Vereczki et al., 2021)
- This group is a minority, estimated at about **5.5-8%** of GABAergic cells in LA/BA. (Vereczki et al., 2021)
- **Summary list for the basolateral/corticobasal amygdala**
- Putting the named cell types together, the BLA and related corticobasal amygdala include: **glutamatergic principal/pyramidal-like neurons; PV basket cells; PV axo-axonic cells; CCK/CB1 basket cells; SOM dendrite-targeting interneurons; calbindin-positive dendrite-targeting interneurons; NPY neurogliaform cells; VIP/calretinin interneuron-selective interneurons; and GABAergic projection neurons**. This supports the view that these nuclei contain most major cortical inhibitory classes plus the dominant excitatory projection neurons. (Vereczki et al., 2016) (Vereczki et al., 2021) (Bienvenu et al., 2012) (Woodruff et al., 2007)

Evidence

(Woodruff et al., 2007)

Networks of Parvalbumin-Positive Interneurons in the Basolateral Amygdala

> *"The main input nucleus of the amygdala is the basolateral amygdala (BLA), a cortical-like structure in which two main types of neuron have been described: glutamatergic principal neurons and GABAergic interneurons (McDonald, 1992;(Smith et al., 1998). Principal neurons, which are similar to pyramidal neurons in the hippocampus and cortex, constitute the majority (85-90%) of the BLA neuronal population (McDonald, 1992)"*

> *"Four populations of interneurons have been described in the BLA: those expressing parvalbumin (McDonald, 1992;Mc-Donald and Betette, 2001), those expressing somatostatin (Mc-Donald and Mascagni, 2002), those expressing cholecystokinin"*

(Washburn et al., 1992)

Electrophysiological and morphological properties of rat basolateral amygdaloid neurons in vitro

> *"The vast majority of cells studied were identified as pyramidal cells"*

> *"The second class of cells consisted of late-firing neurons"*

> *"The third class of cells, termed fast-firing neurons, possessed many of the features of intrinsic inhibitory interneurons found elsewhere in the brain"*

> *"Our data support the conclusion derived from previous anatomical studies that pyramidal neurons constitute the predominant cell type in the BLA and function as projection neurons in this region of the amygdala."*

(Niimi et al., 2012)

Heterogeneous electrophysiological and morphological properties of neurons in the mouse medial amygdala in vitro.

> *"Most studied cells were identified as pyramidal cells"*

> *"The second class of cells consisted of latefiring neurons"*
>
> *"The third class of cells, termed first-firing neurons"*
>
> *"Biocytin labeling of electrophysiologically identified pyramidal and late-firing neurons revealed that the cells have pyramidal to stellate cell bodies"*

(Vereczki et al., 2016)

Synaptic Organization of Perisomatic GABAergic Inputs onto the Principal Cells of the Mouse Basolateral Amygdala

> *"As in other cortical regions, the BLA is composed of glutamatergic principal cells (PCs) and a variety of GABAergic interneurons (McDonald, 1992)(Sah et al., 2003)(Pape et al., 2010)"*
>
> *".We found that the soma and proximal dendrites of PCs were innervated primarily by two neurochemically distinct basket cell types expressing parvalbumin (PVBC) or cholecystokinin and CB1 cannabinoid receptors (CCK/CB1BC). The innervation of the initial segment of PC axons was found to be parceled out by PVBCs and axo-axonic cells (AAC)"*

(Vereczki et al., 2021)

Total Number and Ratio of GABAergic Neuron Types in the Mouse Lateral and Basal Amygdala

> *"we estimated that the following cell types together compose the vast majority of GABAergic cells in the LA and BA: axo-axonic cells (5.5%-6%), basket cells expressing parvalbumin (17%-20%) or cholecystokinin (7%-9%), dendrite-targeting inhibitory cells expressing somatostatin (10%-16%), NPY-containing neurogliaform cells (14%-15%), VIP and/or calretinin-expressing interneuron-selective interneurons (29%-38%), and GABAergic projection neurons expressing somatostatin and neuronal nitric oxide synthase (5.5%-8%). Our results show that these amygdalar nuclei contain all major GABAergic neuron types as found in other cortical regions."*

(Rovira-Esteban et al., 2019)

Excitation of Diverse Classes of Cholecystokinin Interneurons in the Basal Amygdala Facilitates Fear Extinction

> *"One commonly adopted method segregates IN subpopulations based on neurochemical content, including expression of Ca 2+ -binding proteins [e.g., parvalbumin (PV); (McDonald et al., 2001)(McDonald et al., 2001)] and neuropeptides such as somatostatin (SOM), neuropeptide Y (NPY), and cholecystokinin (CCK; Mascagni and Mc-Donald, 2003;(Kepecs et al., 2014)"*
>
> *".four IN types could be identified among the EYFP-expressing cells: CCK/cannabinoid receptor type 1 (CB1R)-expressing basket cells, neurogliaform cells, PV+ basket cells, and PV+ axo-axonic cells."*

(Bienvenu et al., 2012)

Cell-Type-Specific Recruitment of Amygdala Interneurons to Hippocampal Theta Rhythm and Noxious Stimuli In Vivo

> *"We characterized GABAergic interneuron types of the BLA"*
>
> *"calbindin-positive interneurons targeting dendrites"*
>
> *"parvalbumin-positive basket cells"*
>
> *"axo-axonic cells"*
>
> *"a disctinct projecting interneuron type."*

Central amygdala cell types

The central amygdala is mainly made of inhibitory neurons and is usually divided into capsular, lateral, and medial parts. Its most often named cell types are PKC-δ neurons and somatostatin neurons in the lateral/capsular central amygdala, along with older morphology- and firing-based classes such as medium spiny, large aspiny, small aspiny, late-firing, regular-spiking, burst-firing, and fast-spiking neurons. (8 sources)

- **Overall central amygdala neuron class**
- The **central amygdala (CeA)** has a **striatal-like organization** and is composed mostly of **GABAergic inhibitory neurons**, unlike the glutamatergic principal-cell-dominated basolateral nuclei. (Yeh et al., 2024) (Adke et al., 2019) (Gilpin et al., 2014)
- **Major CeA subdivisions**
- Classical descriptions divide the CeA into **capsular (CeC)**, **lateral (CeL)**, and **medial (CeM)** parts. These

subdivisions are used when naming many CeA cell types. (Nisbett et al., 2025)

- **PKC-δ neurons**
- A notable genetically defined CeA cell type is the **protein kinase C-δ-positive (PKC-δ+) neuron**, especially in **CeL/CeC**. PKC-δ and somatostatin cells together make up most neurons in the lateral central amygdala and are largely non-overlapping. (Adke et al., 2019) (Nisbett et al., 2025)
- These cells are commonly linked to the classical **CeL OFF** population and are described as **predominantly late-firing neurons**. (Nisbett et al., 2025) (Haubensak et al., 2010)
- **Somatostatin neurons**
- The other major CeL/CeC class is the **somatostatin-positive (SOM+) neuron**. These cells also form a large share of lateral central amygdala neurons and are largely separate from PKC-δ cells. (Adke et al., 2019) (Nisbett et al., 2025)
- SOM neurons are reported as **late-firing or regular-spiking**, and they are a notable named CeA class in fear-circuit studies. (Nisbett et al., 2025) (Li et al., 2013)
- **CRF neurons**
- Recent rodent work also highlights **CRF neuronal populations** in CeA, alongside PKC-δ and SOM populations. (Yeh et al., 2024)
- **Morphological classes**
- Older anatomical work described several CeA neuron morphologies. One common type in the lateral sector is the **medium spiny neuron**, with an **ovoid soma**, **primary nonspiny dendrites**, **spiny secondary and tertiary dendrites**, and axons that branch before leaving the nucleus. (Nikolenko et al., 2020)
- Other described forms include **large aspiny neurons** with **big somata and thick aspiny dendrites**, and **small aspiny neurons**. (Nikolenko et al., 2020)
- **Electrophysiological classes**
- CeA neurons have also been grouped by firing pattern into **type A** and **type B** cells. In the cited summary, type A cells made up about **75%** and type B cells about **25%**. (Nikolenko et al., 2020)
- Additional firing classes include **burst-firing cells** and **fast-spiking cells**. Fast-spiking cells are described as many of the local internuncial neurons, while burst-firing cells can be split into **group I** and **group II** forms. (Nikolenko et al., 2020)
- **Newer molecularly defined CeA types beyond classical markers**
- Single-cell work argues that classical marker-defined types do **not** fully cover the CeA. It identified **two major cell types** that had not been resolved in earlier schemes, including a **Nr2f2-expressing** population in a non-canonical CeA subdomain and an **Isl1-expressing medial cell type** that accounts for many long-range CeA projections. (O'Leary et al., 2022)
- **Compact summary list for CeA**
- Putting the named CeA cell types together, the central amygdala includes: **PKC-δ+ neurons, SOM+ neurons, CRF neurons, medium spiny neurons, large aspiny neurons, small aspiny neurons, type A cells, type B cells, burst-firing cells, fast-spiking cells, Nr2f2+ cells, and Isl1+ medial projection cells**. (Adke et al., 2019) (Nikolenko et al., 2020) (O'Leary et al., 2022) (Yeh et al., 2024)

Evidence

(Yeh et al., 2024)

Molecular diversity and functional dynamics in the central amygdala

> *"The pallial portion, encompassing the BLA and CoA, exhibits a cortical-like structure predominantly composed of glutamatergic (excitatory) neurons. In contrast, the CeA neurons, originating from the subpallial region, show a striatal-like organization with a majority of GABAergic (inhibitory) neurons (Swanson et al., 1983)(Sah et al., 2003); Figure 1A). The MeA, deriving from both ventral pallial and subpallial origins, presents a diverse neuronal population (Garcia-Lopez et al., 2008;Bupesh et al., 2011)"*

> *".It revealed distinct PKC-δ, SOM, and CRF neuronal populations similar to those in mice."*

(Adke et al., 2019)

Cell-Type Specificity of Neuronal Excitability and Morphology in the Central Amygdala

> *"This diverse span of function is mirrored by the genetic, physiological and morphologic heterogeneity in CeA neuron subtypes (Martina et al., 1999;Schiess et al., 1999;Janak and Tye, 2015).*
>
> *Two genetically identified cell types, protein kinase Cdexpressing (PKCd 1) neurons and somatostatin-expressing (Som 1) neurons, constitute most CeLC neurons and are largely non-overlapping (Li et al., 2013;Kim et al., 2017;Wilson et al., 2019)."*

(Gilpin et al., 2014)

The Central Amygdala as an Integrative Hub for Anxiety and Alcohol Use Disorders

> *"The central amygdala (CeA) plays a central role in physiological and behavioral responses to fearful stimuli, stressful stimuli, and drug-related stimuli. The CeA receives dense inputs from cortical regions, is the major output region of the amygdala, is primarily GABAergic (inhibitory), and expresses high levels of pro- and anti-stress peptides. The CeA is also a constituent region of a conceptual macrostructure called the extended amygdala that is recruited during the transition to alcohol dependence. In this review, we discuss neurotransmission in the CeA as a potential integrative hub between anxiety disorders and Alcohol Use Disorder (AUD), which are commonly co-occurring in humans. Human imaging work and multi-disciplinary work in animals collectively suggest that CeA structure and function are altered in individuals with anxiety disorders and AUD, the end result of which may be disinhibition of downstream "effector" regions that regulate anxiety- and alcohol-related behaviors."*

(Nisbett et al., 2025)

Neuronal Colocalization of μ-Opioid Receptor, κ-Opioid Receptor, and Oxytocin Receptor mRNA in the Central Nucleus of the Amygdala in Male and Female Mice

> *"As discussed above, the central amygdala can be divided into three parts: CeC, CeL, and CeM, based on immunohistochemical markers, morphology, and function (McDonald, 1982)(Cassell et al., 1986)(Jolkkonen et al., 1998)(Davis et al., 2000)Pitkanen, 2000; (Dong et al., 2001)"*

> *".previous studies that categorized cell populations of the CeL (and CeC) based on electrophysiological properties and immunohistochemical markers (Cassell et al., 1989)(Ciocchi et al., 2010)(Haubensak et al., 2010). One population expresses protein kinase Cδ and comprises predominantly late-firing neurons (Li et al., 2013). They are referred as CeL OFF neurons because they respond to fear-conditioned stimuli with reduced activity (Ciocchi et al., 2010)(Haubensak et al., 2010). The other population instead expresses somatostatin, which can be later firing or regular spiking (Li et al., 2013)."*

(Haubensak et al., 2010)

Genetic dissection of an amygdala microcircuit that gates conditioned fear

> *"The role of different amygdala nuclei (neuroanatomical subdivisions) in processing Pavlovian conditioned fear has been studied extensively, but the function of the heterogeneous neuronal subtypes within these nuclei remains poorly understood. Here we use molecular genetic approaches to map the functional connectivity of a subpopulation of GABA-containing neurons, located in the lateral subdivision of the central amygdala (CEl), which express protein kinase C-δ (PKC-δ). Channelrhodopsin-2-assisted circuit mapping in amygdala slices and cell-specific viral tracing indicate that PKC-δ+ neurons inhibit output neurons in the medial central amygdala (CEm), and also make reciprocal inhibitory synapses with PKC-δ− neurons in CEl. Electrical silencing of PKC-δ+ neurons in vivo suggests that they correspond to physiologically identified units that are inhibited by the conditioned stimulus, called CEloff units. This correspondence, together with behavioural data, defines an inhibitory microcircuit in CEl that gates CEm output to control the level of conditioned freezing."*

(Li et al., 2013)

Experience-dependent modification of a central amygdala fear circuit

> *"The amygdala is essential for fear learning and expression. The central amygdala (CeA), once viewed as a passive relay between*

the amygdala complex and downstream fear effectors, has emerged as an active participant in fear conditioning. However, the mechanism by which CeA contributes to the learning and expression of fear is unclear. We found that fear conditioning in mice induced robust plasticity of excitatory synapses onto inhibitory neurons in the lateral subdivision of the CeA (CeL). This experience-dependent plasticity was cell specific, bidirectional and expressed presynaptically by inputs from the lateral amygdala. In particular, preventing synaptic potentiation onto somatostatin-positive neurons impaired fear memory formation. Furthermore, activation of these neurons was necessary for fear memory recall and was sufficient to drive fear responses. Our findings support a model in which fear conditioning–induced synaptic modifications in CeL favor the activation of somatostatin-positive neurons, which inhibit CeL output, thereby disinhibiting the medial subdivision of CeA and releasing fear expression."

(Nikolenko et al., 2020)

Amygdala: Neuroanatomical and Morphophysiological Features in Terms of Neurological and Neurodegenerative Diseases

"Morphologically, there are several types of neurons located in the central nucleus of the amygdala (CeA). In the lateral sector of the central nucleus, a predominant cell type with ovoid soma is located. These cells have several primary nonspiny dendrites, branching onto spiny secondary and tertiary dendrite. Their axons begin branching even before leaving the nucleus, which is why these cells are called "medium spiny neurons" (Hall, 2004)(McDonald, 1982). Another type of neurons located in the central nuclei have big soma with thick aspiny dendrites, branching on to secondary seldom spiny processes (McDonald, 1982)(Cassell et al., 1989)(Schiess et al., 1999). The third type of cells are small aspiny neurons (Cassell et al., 1989)"

".Two distinct cell types were described in terms of their reaction to a prolonged and short current injection: type A (75%) and type B (25%) (Schiess et al., 1999)(Schiess et al., 1993). Another two types of cells called "burst-firing" cells and "fast-spiking" cells are present in the amygdala. Most of the internuncial neurons are fast-spiking cells showing no adaptation and high frequency of rapid action potential (Martina et al., 1999). Burst-firing cells can be subdivided into two groups: group I has adaptation and shows recommencing bursts of action potentials (making them alike to the B-type cells described above), and group II has neurons with low-threshold burst activity, which are similar to type A cells, described above (Schiess et al., 1993)(Martina et al., 1999)."

(O'Leary et al., 2022)

Neuronal cell types, projections, and spatial organization of the central amygdala

"Such work has typically defined molecular cell types by classical inhibitory marker genes; consequently, whether marker-gene-defined cell types exhaustively cover the CEA and co-vary with connectivity remains unresolved. Here, we combined single-cell RNA sequencing, multiplexed fluorescent in situ hybridization, immunohistochemistry, and long-range projection mapping to derive a "bottom-up" understanding of CEA cell types. In doing so, we identify two major cell types, encompassing one-third of all CEA neurons, that have gone unresolved in previous studies. In spatially mapping these novel types, we identify a non-canonical CEA subdomain associated with Nr2f2 expression and uncover an Isl1-expressing medial cell type that accounts for many long-range CEA projections."

Medial, cortical/superficial, and intercalated cell populations

Outside the basolateral and central nuclei, the amygdala also includes medial, cortical/superficial, and intercalated populations that add important cell-type diversity. In broad terms, the superficial/cortical nuclei are mainly cortical-like excitatory populations, the medial amygdala is mostly inhibitory but mixed, and the intercalated cell masses are specialized clusters of GABAergic interneurons. (3 sources)

Beyond the better-known basolateral and central nuclei, the remaining amygdala contains several additional classical and molecularly defined cell populations. The **cortical or superficial amygdala** is generally grouped with the cortical-like side of the amygdala and is therefore expected to contain mainly **glutamatergic projection neurons** together with local inhibitory cells, rather than the mostly GABAergic organization seen in central amygdala (Model-Generated). This same broad cortical-like logic extends to the **basomedial/anterior-intermediate region** in newer developmental and spatial maps, where the **anterior intermediate stratum** corresponds to the region classically called the **basomedial amygdala (BMA)**; in that framework, multiple annotated cell types were distinguished in this anterior radial cluster, including **Vwa5b1/Zbtb7c, Vwa5b1/Sim1, Rorb, Tfap2c, Abi3bp, and Satb2/Ebf2** populations (Fernandez et al., 2025).

The **medial amygdala (MeA)** sits somewhat between the cortical-like and striatal-like poles in classical descriptions. In broad neurochemical terms it is usually described as **predominantly GABAergic**, but unlike the central nucleus it is not purely so, because rodent studies also report **glutamatergic projection neurons** there (Model-Generated). Classical accounts therefore usually treat the medial nucleus as a mixed but mostly inhibitory population rather than as a simple excitatory principal-cell field (Model-Generated).

The **intercalated cell masses (ITCs)** are the clearest additional named population. These are small groups of densely packed neurons located between the major nuclei, especially between basolateral territories and the central nucleus. Recent human and primate-oriented transcriptomic work specifically identified **distinct subtypes of FOXP2+ interneurons in the intercalated cell masses**, reinforcing the view that ITCs are a specialized inhibitory population rather than just an extension of neighboring nuclei (Totty et al., 2024). The same study also emphasizes that major amygdala nuclei contain both **glutamatergic and GABAergic neurons** across species and that molecular specialization can be resolved at the level of individual nuclei and cell classes (Totty et al., 2024) (McDonald, 2020).

So, for the user's "all cell types" question, the main additions in these regions are: **cortical/superficial cortical-like excitatory neurons with local interneurons; medial-amygdala mostly GABAergic neurons plus some glutamatergic projection neurons; and intercalated FOXP2+ GABAergic interneuron populations**. In newer spatial and transcriptomic schemes, these broad classical groups can be split further into region-linked molecular populations such as the **Vwa5b1/Zbtb7c, Vwa5b1/Sim1, Rorb, Tfap2c, Abi3bp, and Satb2/Ebf2** cell classes mapped to the anterior intermediate/BMA-related territory (Fernandez et al., 2025).

Evidence

(Fernandez et al., 2025)

Transcriptomic Analysis Corroborates the New Radial Model of the Mouse Pallial Amygdala

> *"In the anterior radial cluster, Yu et al. [14] annotated six cell types (Vwa5b1/Zbtb7c; Vwa5b1/Sim1; Rorb; Tfap2c; Abi3bp; and Satb2/Ebf2)"*

> *".we also add the corresponding ant-i stratum, classically known as BMA. In addition, our clusters for the lateral, basal, and posterior radial domains were jointly identified as 'the BLA area' in the Yu et al. [14] dataset."*

(Totty et al., 2024)

Transcriptomic diversity of amygdalar subdivisions across humans and nonhuman primates

> *"The majority of amygdala neurons are located in the lateral, basal, accessory basal, intercalated and central nuclei"*

> *". Targeting specific nuclei with neuronal tracers in combination with morphological analyses in the nonhuman primate, has provided clear descriptions of the connectivity of glutamatergic and GABAergic neurons across the major amygdalar nuclei (McDonald, 2020) (Cho et al., 2013)(McHale et al., 2021)"*

> *"We identified distinct subtypes of FOXP2+ interneurons in the intercalated cell masses and protein-kinase C-δ interneurons in the central nucleus. We also establish that glutamatergic, pyramidal-like neurons are transcriptionally specialized within the basal, lateral, or accessory basal nuclei"*

> *". Multiple classes of GABAergic interneurons, particularly those expressing LAMP5 and SST, were enriched for genes downregulated in postmortem brain tissue of individuals diagnosed with post-traumatic stress disorder (PTSD) 27 ."*

(McDonald, 2020)

Functional neuroanatomy of the basolateral amygdala: Neurons, neurotransmitters, and circuits

> *"Abstract The basolateral nuclear complex (BNC) of the amygdala consists of lateral, basolateral, and basomedial nuclei in the rat, and homologous nuclei in the monkey (lateral, basal, and accessory basal). There are two main types of neurons in the BNC: pyramidal neurons and nonpyramidal neurons. Although these cells do not exhibit a laminar organization, their morphology, synaptology, electrophysiology, and pharmacology are remarkably similar to their counterparts in the cerebral cortex. Like the cortex, there are several distinct interneuronal subpopulations in the BNC. Collectively, the nuclei of the BNC receive information from all sensory modalities, and via projections to the central amygdalar nucleus, bed nucleus of the stria terminalis, striatum, hypothalamus, and prefrontal cortex elicit emotional behavior. The BNC also plays a critical role in the formation and extinction of emotional memories. This review summarizes the organization of the local and extrinsic connections of BNC neurons that underly these functions."*

Cross-species and whole-amygdala molecular taxonomy in humans and rodents

Whole-amygdala single-cell studies show that the amygdala contains many more molecularly distinct cell types than the older classical schemes suggested. Across mouse, rodent, primate, and human datasets, the same broad classes recur: excitatory neurons, inhibitory neurons, and a full set of non-neuronal support cells, while the exact subtype counts differ by study and species. (7 sources)

Recent whole-amygdala transcriptomic work adds an important final layer to the classical picture above: the amygdala is not just made of a few named neuron classes, but of many molecularly separable subtypes spread across its nuclei. In mouse, a full-amygdala single-cell atlas identified **130 neuronal cell types** and mapped them across the structure, showing that fear-related amygdala circuits are built from a very large number of transcriptionally distinct neuronal populations rather than only the classical projection-neuron/ interneuron split. (Hochgerner et al., 2022) A later mouse taxonomy grouped these neuronal populations into **three main molecular neuronal classes**: **56 GABAergic types**, **32 VGLUT1 excitatory types**, and **42 VGLUT2 excitatory types**, which fits well with the known mix of cortical-like and striatal-like amygdala territories. (Hochgerner et al., 2023) (Pitkanen et al., 1997)

Cross-species studies show that this neuronal diversity sits within a broader shared cellular framework. In four-species comparisons, all major brain cell classes were found in the amygdala, including **excitatory neurons, inhibitory neurons, astrocytes, oligodendrocytes, OPCs, ependymal cells, microglia/macrophages, endothelial cells, and mural cells**. (Yu et al., 2023) This means that a complete answer to "all cell types in the amygdala" should include not only neurons but also the major non-neuronal populations that support vascular, immune, myelin, and homeostatic functions. A recent human amygdala atlas made this explicit by grouping cells into **five broad classes: excitatory neurons, inhibitory neurons, glial cells, stromal cells, and immune cells**, and reported that glia make up about half of all amygdala cells, while immune and stromal cells are sparse. (Gui et al., 2025)

These whole-amygdala datasets also sharpen the cross-nucleus pattern already outlined above. In integrated analyses of BLA and CeA, **CeA clusters cocluster mainly with inhibitory neurons**, whereas **BLA clusters cocluster mainly with excitatory neurons**, while glial populations are shared across subregions rather than being unique to one nucleus. (Zhou et al., 2023) (Beyeler et al., 2020) In humans, the overall balance may differ somewhat from cortex: one recent study estimated an **excitatory-to-inhibitory ratio of about 1.3** in amygdala, still with more excitatory than inhibitory neurons overall, but much less skewed than in cortex. (Gui et al., 2025)

So, at the whole-amygdala and cross-species level, the most complete compact answer is: **many excitatory neuronal subtypes, many inhibitory neuronal subtypes, plus astrocytes, oligodendrocytes, OPCs, microglia/ macrophages, endothelial cells, mural cells, ependymal cells, and small stromal/immune populations**. The exact subtype names and counts change by species and by atlas, but the broad cellular plan is conserved across rodents and humans, while molecular methods now split each classical group into dozens of finer cell types. (Yu et al., 2023) (Hochgerner et al., 2022) (Hochgerner et al., 2023)

Evidence

(Hochgerner et al., 2022)

Cell types in the mouse amygdala and their transcriptional response to fear conditioning

> *"The amygdala is one of the most widely studied regions in behavioral neuroscience. A plethora of classical, and new paradigms have dissected its precise involvement in emotional and social sensing, learning, and memory"*

> *"we present a molecular cell type taxonomy of the full mouse amygdala in fear learning and consolidation. We performed single-cell RNA-seq on naïve and fear conditioned mice, inferred the 130 neuronal cell types distributions in silico using orthogonal spatial transcriptomic datasets"*

(Hochgerner et al., 2023)

Neuronal types in the mouse amygdala and their transcriptional response to fear conditioning

> *"The amygdala consists of multiple anatomical regions, developmentally attributed to the striatal, cortical (superficial) and*

subcortical (deep) structures (Pitkänen et al., 1997)"

". The neuronal cell-type taxonomy describes the following three main classes: 56 GABAergic cell types (Gad1, Gad2 and Slc32a1, 13,006 cells), 32 VGLUT1 types (Slc17a7, 11,947 cells) and 42 VGLUT2 types (Slc17a6, 5,232 cells)."

(Pitkanen et al., 1997)

Organization of intra-amygdaloid circuitries in the rat: an emerging framework for understanding functions of the amygdala.

"The amygdala is located in the medial aspects of the temporal lobe. In spite of the fact that the amygdala has been implicated in a variety of functions, ranging from attention to memory to emotion, it has not attracted neuroscientists to the same extent as its laminated neighbours, in particular the hippocampus and surrounding cortex. However, recently, principles of information processing within the amygdala, particularly in the rat, have begun to emerge from anatomical, physiological and behavioral studies. These findings suggest that after the stimulus enters the amygdala, the highly organized intra-amygdaloid circuitries provide a pathway by which the representation of a stimulus becomes distributed in parallel to various amygdaloid nuclei. As a consequence, the stimulus representation may become modulated by different functional systems, such as those mediating memories from past experience or knowledge about ongoing homeostatic states. The amygdaloid output nuclei, especially the central nucleus, receive convergent information from several other amygdaloid regions and generate behavioral responses that presumably reflect the sum of neuronal activity produced by different amygdaloid nuclei."

(Yu et al., 2023)

Molecular and cellular evolution of the amygdala across species analyzed by single-nucleus transcriptome profiling

"In all four species, all major brain cell classes were identified according to canonical cell-type marker genes, including excitatory neurons, inhibitory neurons, astrocytes, oligodendrocytes, oligodendrocyte progenitor cells (OPC), ependymal cells, microglia/ macrophages, endothelial cells, and mural cells (Supplementary Fig. S1d, e)."

(Gui et al., 2025)

The left amygdala is genetically sexually-dimorphic: multi-omics analysis of structural MRI volumes

"The 19 cell types correspond to five cell groups, including glial, stromal, immune, excitatory and inhibitory neuronal cells"

".The excitatory-to-inhibitory neuron ratio was 1.3 compared to that in human cortex (2.3) and mice (3.0) (Fang et al., 2022), but the direction is the same, with more excitatory neurons than inhibitory neurons. Glial cells constitute approximately half of the total cell population within the amygdala. The presence of immune cells and stromal cells was minimal within the amygdala."

(Zhou et al., 2023)

Single-nucleus genomics in outbred rats with divergent cocaine addiction-like behaviors reveals changes in amygdala GABAergic inhibition

"For most downstream analyses, we focused on the six most common main cell types (Fig. 2a,b)"

".Consistent with the known cell-type composition of the CeA and BLA (Beyeler et al., 2020) , cell clusters from the CeA coclustered primarily with inhibitory neurons whereas those from the BLA coclustered with excitatory neurons"

"Glial cell types from the whole amygdala contained cells from both subregions, except for astrocytes, which coclustered mostly with cells from the CeA but not those from the BLA"

(Beyeler et al., 2020)

Neuronal diversity of the amygdala and the bed nucleus of the stria terminalis

"Abstract The amygdala complex is a diverse group of more than 13 nuclei, segregated in five major groups: the basolateral (BLA), central (CeA), medial (MeA), cortical (CoA), and basomedial (BMA) amygdala nuclei. These nuclei can be distinguished depending on their cytoarchitectonic properties, connectivity, genetic, and molecular identity, and most importantly, on their functional role in animal behavior. The extended amygdala includes the CeA and the bed nucleus of the stria terminalis (BNST). Both CeA and the BNST share similar cellular organization, including common neuron types, reciprocal connectivity, and many overlapping downstream targets. In this section, we describe the advances of our knowledge on neuronal diversity in the amygdala complex and the BNST, based on recent functional studies, performed at genetic, molecular, physiological, and anatomical levels in rodent models, especially rats and mice. Molecular and connection property can be used separately, or in combinations, to define neuronal populations, leading to a multiplexed neuronal diversity-supporting different functional roles."